

CS160 Group 6 – Sprint Plan | Due: April 27, 20

Task Name	Description	Priority	Est. Days	Owner
Week 1 (Mar 2–4) — Scrum Review on Wednesday				
Code Review Checklist	Define pre-merge workflow; another teammate reviews code/tests before GitHub merge.	Medium	1	P1
Finalize DB Schema & Tables	Create/update MySQL schema for users, products, cart, cart_items, orders, order_items, robots, schedules, routes, status_history.	High	3	P2
Seed Sample Product/User Data	Populate realistic demo/test data for products, users, admins, inventory, and sample orders.	Medium	2	P3
Week 2 (Mar 5–11) — Scrum Review on Wednesday				
Auth API – Register	Implement POST /api/auth/register with validation, secure password hashing, and user creation.	High	3	P4
Auth API – Login/Logout	Implement FastAPI auth endpoints for login/logout, JWT issuance, cookie handling, and logout cookie clearing.	High	3	P5
Auth API – Me Endpoint	Implement GET /api/auth/me to return current authenticated user profile and role.	High	2	P6
Auth Middleware / Require Auth	Add backend auth guard for protected routes and role checks for customer/admin access.	High	2	P1
Frontend Login Page	Build React login form, validation, error handling, role-based redirect on success.	High	2	P2
Session Validation / Role Routing	Add frontend logic to fetch current session, resolve role, and route user/admin appropriately.	High	2	P3
Frontend Registration Page	Build registration UI with validation and success/error handling; connect to register API.	High	3	P4
Week 3 (Mar 12–18) — Scrum Review on Wednesday				
Product List API	Implement /api/products with search, filter, sort, pagination, and deterministic responses.	High	4	P5
Product Browsing UI	Build product listing page with search/filter UI and dynamic re-render on filter change.	High	4	P6
Product Detail API	Implement /api/products/{id} endpoint to return product details and stock info.	High	2	P1
Product Detail UI	Build product detail page with quantity validation and Add-to-Cart action.	High	3	P2
Unit Tests – Auth ♦	Write pytest unit tests for register, login, /me, and middleware role checks.	High	2	P3
Week 4 (Mar 19–25) — Scrum Review on Wednesday				
Cart Add/Update/Delete APIs	Implement cart endpoints for add item, update quantity, remove item, and fetch cart state.	High	4	P4
Shopping Cart UI	Build cart page showing items, quantity updates, remove actions, and running totals.	High	4	P5
Delivery Fee Calculation Logic	Implement backend delivery fee rule: under 20 lbs free, otherwise \$10.	High	2	P6

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Task Name	Description	Priority	Est. Days	Owner
Cart Summary API	Implement /api/cart/summary returning items, total price, total weight, and delivery fee.	High	1	P1
Mock Payment Processing	Add mock payment validation behavior for checkout (no real payment integration).	Medium	2	P2
Unit Tests – Products & Cart ♦	Write unit tests for product search/filter helpers, cart validation, pricing/delivery logic.	High	2	P3
Week 5 (Mar 26–Apr 1) — Scrum Review on Wednesday				
Checkout API	Implement /api/checkout flow: validate cart/stock, create order, decrement inventory, clear cart.	High	5	P4
Checkout UI	Build checkout page with delivery date/time selection, payment fields, validation, and confirmation redirect.	High	4	P5
Concurrency / Low-Stock Handling	Add backend safeguards for simultaneous checkout on low-stock items.	High	3	P6
Order Creation + Persistence	Implement order and order_items persistence with snapshot data saved at checkout time.	High	2	P1
Admin Dashboard Auth + Shell	Build admin dashboard layout, session validation, and role-gated navigation.	High	3	P2
Week 6 (Apr 2–8) — Scrum Review on Wednesday				
Order History API	Implement customer endpoint to fetch past orders and details.	High	2	P3
Order History UI	Build order history page so users can view previously placed orders.	Medium	3	P4
Delivery Scheduling Service	Implement scheduling logic that batches/assigns robot trips under constraints (<=10 orders, <=200 lbs).	High	5	P5
Admin Inventory APIs	Implement admin endpoints to view, add, edit, delete products and update stock.	High	4	P6
Admin Inventory UI	Build admin product/inventory management screens with validation and feedback.	High	5	P1
Week 7 (Apr 9–15) — Scrum Review on Wednesday				
Route Optimization Integration	Integrate Google Maps/Map wrapper to generate route and ETA for delivery trips.	High	5	P2
Order Status / Location APIs	Implement GET /api/orders/{id}/status and GET /api/orders/{id}/location.	High	3	P3
Admin/Robot Order Status Update API	Implement PATCH /admin/orders/{id}/status with valid state-transition rules.	High	2	P4
Admin Orders View	Build admin orders view with filtering by status/date/customer and summary reporting.	Medium	3	P5
Revenue / Reports Endpoint	Implement backend endpoint for manager/admin revenue reports used by dashboard.	Medium	2	P6

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Task Name	Description	Priority	Est. Days	Owner
Alpha User Testing	Conduct observed testing with small set of users on core tasks to identify usability issues early.	Medium	2	P1
Week 8 (Apr 16–22) — Scrum Review on Wednesday				
Order Tracking UI + Map	Build order tracking page with polling, ETA, live status text, map marker, and route polyline.	High	6	P2
Test Framework Setup + Seeded Test DB	Configure Jest, pytest, and a dedicated seeded test database for repeatable automated tests.	High	3	P3
Unit Tests – Frontend + Backend	Create unit tests for auth, search/filter helpers, cart validation, pricing logic, scheduling, and status transitions.	High	4	P4
Feature / Integration Tests	Test registration/login, product browsing, cart flow, checkout, inventory update, admin access, and status APIs.	High	4	P5
Frontend Error States + Validation Pass	Add consistent error states for invalid inputs, out-of-stock, auth failure, and network/API failure.	Medium	3	P6
E2E Smoke Test – Happy Path ♦	Run full end-to-end test: register -> browse -> add to cart -> checkout -> track order.	High	2	P1
Week 9 (Apr 23–27) — Scrum Review on Wednesday				
Cross-browser / Environment Testing	Validate required flows in Chrome and Firefox.	Medium	2	P2
Performance + Degraded Network Testing	Measure responsiveness of core flows and verify graceful failure/recovery under temporary connectivity loss.	Medium	2	P3
Load Testing	Simulate multiple concurrent users, monitor stability/response time, and look for race conditions.	Medium	2	P4
Beta User Testing	Conduct near-complete build testing in realistic conditions and collect usability/error feedback.	Medium	2	P5
Final Bug Fix + Regression Pass	Resolve defects from integration/system/load testing and rerun core regression flows.	High	2	P6
Accessibility Audit ♦	Check keyboard navigation, contrast ratios, and ARIA labels on key pages.	Medium	1	P1
Documentation + Demo Prep	Finalize docs, screenshots, API notes, speaking/demo flow, and presentation readiness.	High	2	P2
API Documentation ♦	Write/generate Swagger or README docs for all public API endpoints.	Medium	1	P3
Final Demo Dry-Run ♦	Full walkthrough of demo script as a team; fix any last presentation issues.	Medium	1	P4

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Sprint Week	Start Date	End Date	Done ?	Notes / Comments
Week 1 (Mar 2–4)	03/02/26	03/02/26	☐	Added to match final HLTP
Week 1 (Mar 2–4)	03/02/26	03/04/26	☐	Include FKs, indexes, and order-history support
Week 1 (Mar 2–4)	03/02/26	03/03/26	☐	Required by project/testing setup
Week 2 (Mar 5–11)	03/05/26	03/07/26	☐	Added to match final HLTP
Week 2 (Mar 5–11)	03/05/26	03/07/26	☐	HttpOnly cookie flow
Week 2 (Mar 5–11)	03/08/26	03/09/26	☐	Needed for session/role resolution
Week 2 (Mar 5–11)	03/08/26	03/09/26	☐	Supports protected pages/endpoints
Week 2 (Mar 5–11)	03/08/26	03/09/26	☐	Axios + JWT cookie flow
Week 2 (Mar 5–11)	03/10/26	03/11/26	☐	Uses /api/auth/me
Week 2 (Mar 5–11)	03/10/26	03/12/26	☐	Added to match final HLTP
Week 3 (Mar 12–18)	03/10/26	03/13/26	☐	Validate query params
Week 3 (Mar 12–18)	03/12/26	03/16/26	☐	Includes category browsing
Week 3 (Mar 12–18)	03/14/26	03/15/26	☐	Return 404 for invalid product
Week 3 (Mar 12–18)	03/16/26	03/18/26	☐	Handle out-of-stock display
Week 3 (Mar 12–18)	03/17/26	03/18/26	☐	NEW – test coverage for auth layer
Week 4 (Mar 19–25)	03/16/26	03/20/26	☐	Validate quantity and stock bounds
Week 4 (Mar 19–25)	03/18/26	03/22/26	☐	Recalculate totals immediately
Week 4 (Mar 19–25)	03/21/26	03/22/26	☐	Match final project/HLTP rule

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Sprint Week	Start Date	End Date	Done ?	Notes / Comments
Week 4 (Mar 19–25)	03/23/26	03/23/26	☐	Used by checkout page
Week 4 (Mar 19–25)	03/23/26	03/24/26	☐	Per test plan assumptions
Week 4 (Mar 19–25)	03/24/26	03/25/26	☐	NEW – covers product + cart layers
Week 5 (Mar 26–Apr 1)	03/24/26	03/29/26	☐	Use DB transaction to avoid race issues
Week 5 (Mar 26–Apr 1)	03/24/26	03/28/26	☐	Clear payment state after submit
Week 5 (Mar 26–Apr 1)	04/02/26	04/04/26	☐	Locking / transaction checks
Week 5 (Mar 26–Apr 1)	03/30/26	03/31/26	☐	Supports history + tracking
Week 5 (Mar 26–Apr 1)	03/30/26	04/01/26	☐	Products / inventory / orders / reports
Week 6 (Apr 2–8)	04/01/26	04/02/26	☐	Added to match final HLTP E2E
Week 6 (Apr 2–8)	04/01/26	04/03/26	☐	Includes recent order confirmation path
Week 6 (Apr 2–8)	03/30/26	04/04/26	☐	Simulated robots only
Week 6 (Apr 2–8)	04/02/26	04/06/26	☐	Admin-only protected routes
Week 6 (Apr 2–8)	04/03/26	04/08/26	☐	Add/edit/remove/correct mistakes
Week 7 (Apr 9–15)	04/01/26	04/06/26	☐	Save route + ETA in repository
Week 7 (Apr 9–15)	04/05/26	04/07/26	☐	Return status, coordinates, ETA, route
Week 7 (Apr 9–15)	04/05/26	04/06/26	☐	Added to match final HLTP
Week 7 (Apr 9–15)	04/07/26	04/09/26	☐	Matches final LLD
Week 7 (Apr 9–15)	04/07/26	04/08/26	☐	Supports reports view in LLD

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Sprint Week	Start Date	End Date	Done ?	Notes / Comments
Week 7 (Apr 9–15)	04/14/26	04/15/26	☐	Search, cart, checkout focus
Week 8 (Apr 16–22)	04/07/26	04/12/26	☐	Stop polling when delivered
Week 8 (Apr 16–22)	04/09/26	04/11/26	☐	Added to match final HLTP
Week 8 (Apr 16–22)	04/12/26	04/16/26	☐	Based on final HLTP coverage
Week 8 (Apr 16–22)	04/12/26	04/16/26	☐	Includes API + DB interactions
Week 8 (Apr 16–22)	04/09/26	04/11/26	☐	Covers error cases in HLTP
Week 8 (Apr 16–22)	04/15/26	04/16/26	☐	NEW – verifies complete user journey
Week 9 (Apr 23–27)	04/17/26	04/18/26	☐	Required supported browsers
Week 9 (Apr 23–27)	04/17/26	04/18/26	☐	Added to match final HLTP
Week 9 (Apr 23–27)	04/17/26	04/18/26	☐	Added to match final HLTP
Week 9 (Apr 23–27)	04/19/26	04/20/26	☐	Added to match final HLTP
Week 9 (Apr 23–27)	04/19/26	04/20/26	☐	Freeze scope before packaging
Week 9 (Apr 23–27)	04/20/26	04/20/26	☐	NEW – quick accessibility sweep before demo
Week 9 (Apr 23–27)	04/21/26	04/22/26	☐	Ready before presentation
Week 9 (Apr 23–27)	04/21/26	04/21/26	☐	NEW – helps graders/reviewers understand API
Week 9 (Apr 23–27)	04/25/26	04/25/26	☐	NEW – practice run before Apr 27 deadline